



Electrical Installation

System overview and configuration of CAN nodes

BlueVision_Advanced / NewGeneration

Installation and initial operation

Before using this Quick Reference, read and observe the detailed MTU overall documentation on BlueVision_Advanced / NewGeneration!
In particular, all safety instructions in the overall documentation must be observed!



Legend:

P1 n-Engine
P13 n-Shaft
P5 Lube oil
P6 t-Coolant

P14 t-Gear oil
P15 p-Gear control oil
Connector pin (P)
Connector socket (S)

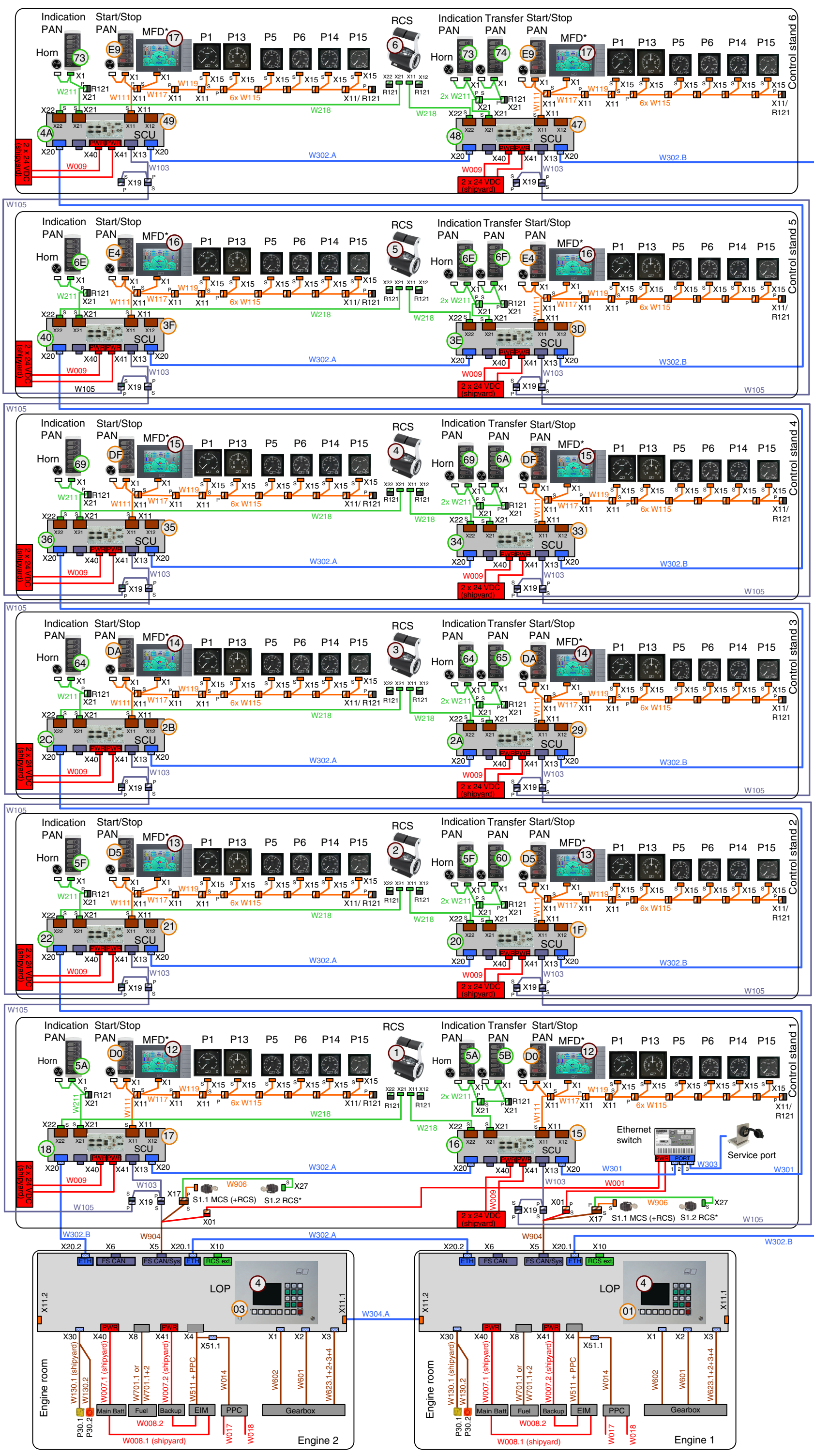
P30.1 Flashing lamp
P30.2 Horn
Option

Notes:

Node numbers are represented in the system wiring as follows:

- ⊗ MCS [hexadecimal]
- ⊗ RCS [hexadecimal]
- ⊗ LOS4, ROS, MFD

System wiring





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Configuration of CAN nodes

Node numbers - control panel LOP (engine room):

Device	Shaft 2	Shaft 4	Shaft 3	Shaft 1	NOTE: At display of LOS4, set the "BDM function" to "Active".
SPU-MCS module	3dec=03hex	7dec=07hex	5dec=05hex	1dec=01hex	
LOS 4 (display)	4 250 kBaud	4 250 kBaud	4 250 kBaud	4 250 kBaud	

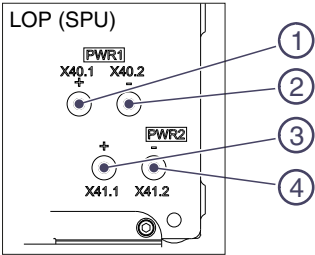
Connection of voltage supply to LOP (shipyard)

Pos.	Connec.pin	Signal	Channel
1	X40.1**	24V_1	Main voltage supply
2	X40.2**	GND_1*	
3	X41.1***	24V_2	Redundant voltage supply
4	X41.2***	GND_2*	

* GND_1 and GND_2 are connected with one another device-internally

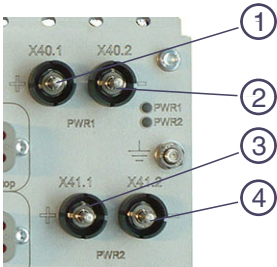
** Cable W007.1 for main voltage supply (provided by shipyard, 10 mm²)

*** Cable W007.2 for redundant voltage supply (provided by shipyard, 10 mm²)

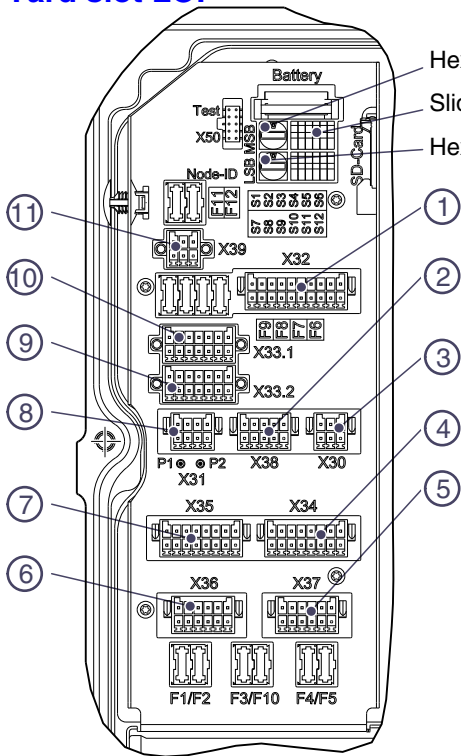


Connection of voltage supply to SCU

Pos.	Connec.	Signal	Channel
1	X40.1	24V_1	Main voltage supply
2	X40.2	GND_1	
3	X41.1	24V_2	Redundant voltage supply
4	X41.2	GND_2	



Yard slot LOP



Hex rotary coding switch MSB

Slide switch

Hex rotary coding switch LSB

Connector assignment

Pos.	Connec.	Designation
1	X32	Non-MTU RCS (FRCS)
2	X38	Non-MTU RCS (FRCS)
3	X30	Horn / flashing lamp
4	X34	Opto 2
5	X37	Relay 2
6	X36	Relay 1
7	X35	Opto 1
8	X31	NMEA (CAN5/RS422)
9	X33.2	Jumper MTU RCS active
10	X33.1	Jumper FRCS active
11	X39	LOS (SPU-internal)

Settings in yard slot of LOP

Position of slide switches

The assignment of the function (e.g. Enginewise-FB Cfg1) and the switch position depends on the software configuration (Automation Suite / Configurator)!

Switch	Yard slot designation	Position	Function
S1*	Enginewise-FB Cfg1	top	Gearbox has no engine-wise feedback, feedback is generated internally in the LOP.
		bottom	Enginewise feedback comes direct from the gearbox.
S2**	Enginewise-FB Cfg2	top	Enginewise feedback switches to +24V (high side switch).
S3**	Enginewise-FB Cfg3	bottom	
S2	Enginewise-FB Cfg2	bottom	Enginewise feedback switches to GND (low side switch.)
S3	Enginewise-FB Cfg3	top	
S4*	Neutral-FB Cfg1	top	Gearbox has no neutral feedback, feedback is generated internally in the LOP.
		bottom	Neutral feedback comes direct from the gearbox.
S5**	Neutral-FB Cfg2	top	Neutral feedback switches to +24V (high side switch).
S6**	Neutral-FB Cfg3	bottom	
S5	Neutral-FB Cfg2	bottom	Neutral feedback switches to GND (low side switch).
S6	Neutral-FB Cfg3	top	
S7*	CountEnginewise-FB Cfg1	top	Gearbox has no counter-engine-wise feedback, feedback is generated internally in the LOP.
		bottom	Counter-engine-wise feedback comes direct from the gearbox.
S8**	CountEnginewise-FB Cfg2	top	Counter-engine-wise feedback switches to +24V (high side switch).
S9**	CountEnginewise-FB Cfg3	bottom	
S8	CountEnginewise-FB Cfg2	bottom	Counter-engine-wise feedback switches to GND (low side switch).
S9	CountEnginewise-FB Cfg3	top	
S11	Keyswitch Master	top	Keyswitch master not available: Plant switches on automatically when voltage supply is applied.
		bottom	Plant is switched on via the external Keyswitch Master.
S12	Keyswitch RCS	top	Keyswitch RCS not available: Plant switches on automatically when voltage supply is applied.
		bottom	Keyswitch RCS switches the RCS control stand components in the SPU (CAN) and RCS-IO extension on and off.

*The position of the slide switches S1, S4, S7 influences the output at the shipyard interface (in each case, the function depends on the software configuration):

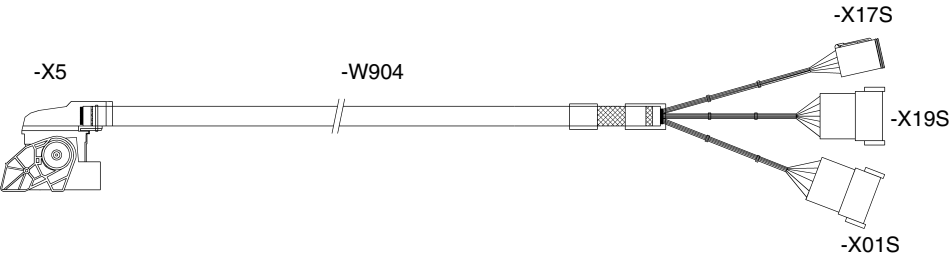
S1: -X36 pin 7, 9, 11

S4: -X36 pin 1, 3, 5

S7: -X36 pin 2, 4, 6

**S2 and S3, S5 and S6, S8 and S9 are always connected in combination.

Connection, engine room (LOP) to control stand 1

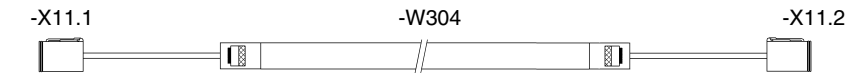


from	to	Conduc. no.	Conduc. color	Signal	from	to	Conduc. no.	Conductor color	Signal
-X5	-X01S	1	white	+24V	A5	1	5	gray	KS MCS_+24V
C1	2	2	brown	GND	D5	4	6	pink	KS RCS_+24V
B1					B6	2	7	blue	KS MCS_OFF
					B7	3	8	red	KS MCS_ON
					C6	5	9	black	KS RCS_OFF
					C7	6	10	purple	KS RCS_ON
								GNYE	Shield

from	to	Conduc. no.	Conduc. color	Signal	from	to	Conduc. no.	Conductor color	Signal
-X5	-X19S	1	green	EM_Stop_H	A1				
A8	2	4	yellow	EM_Stop_L					
D8									

Cable designation: XZ00E50000694

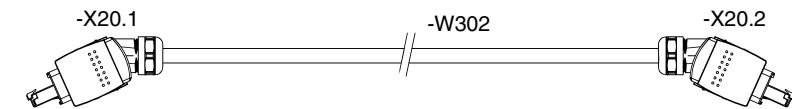
Connection "Network On" between the LOPs of the power trains



from	to	Conduc. no.	Conduc. color	Signal
-X11.1	-X11.2	1	white	Switch on network
1	2	2	brown	Switch on network
2	4	4	GNYE	Shield

Cable designation: XZ00E50000767

Connection of system bus between the LOPs, between the SCUs and to the SCU FST 1 / last control stand



from	to	Conduc. color	Signal
-X20.1	-X20.2	YE	TD+
1	2	OG	TD-
2	3	WH	RD+
3	6	BU	RD-
6			Shield
Heat shield	Heat shield		

Cable designation: XZ00E50000696